

A4 Machine Learning

Student Workbook

IB DP Computer Science 2027

Name: _____ Class: _____

Date: _____ SL / HL (circle one)

How to Use This Workbook

This workbook accompanies the **A4 ML Projects** suite. For each project:

1. Open the HTML file in your browser
2. Read the **real Python code** shown
3. Look at the **real output** from the actual data
4. Click  **Run in Colab** to experiment yourself
5. Answer the questions below

Datasets used (all real):

- California Housing (US Census, 20,640 houses)
- Telco Customer Churn (Kaggle, 7,043 customers)
- Mall Customers (200 shoppers)
- Online Retail (5,000 real transactions)
- Fashion-MNIST (10,000 clothing images)

Project 1: EDA & ML Types

Key Concepts

Term	Definition
Supervised Learning	Model learns from labelled data (features → target)
Unsupervised Learning	Model finds patterns in unlabelled data
Reinforcement Learning	Agent learns through trial-and-error + reward
Regression	Predicts a continuous numerical value
EDA	Exploratory Data Analysis — understand data through visualization

Questions

1. Look at the California housing data. What type of ML problem is this? Why?

Answer:

2. The correlation heatmap shows `MedInc` (median income) has the strongest correlation with house value. What does this tell you?

Project 2: Data Preprocessing

Key Concepts

Term	Definition
Missing Value	Data point that doesn't exist (NaN)
Outlier	Data point far from the normal range
StandardScaler	Transforms data to mean=0, std=1
MinMaxScaler	Transforms data to range [0, 1]
PCA	Principal Component Analysis — reduces dimensions
Feature Selection	Choosing which features are most useful

Questions

1. What were the three types of data quality problems injected into the dataset?

Answer:

2. The IQR method uses $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$ as fences. What percentage of normally distributed data falls outside these fences? (Hint: think about standard deviations)

Project 3: Linear Regression

Key Concepts

Term	Definition
R ² (R-squared)	Proportion of variance explained by the model (0 to 1)
RMSE	Root Mean Squared Error — average prediction error
Residual	Actual value - Predicted value
Coefficient	Weight assigned to each feature
Regularization	Penalty to prevent overfitting (Ridge/Lasso)

Questions

1. Compare the R² values: Simple regression (MedInc only) vs Multiple regression (8 features). Which is better and by how much?

Answer:

2. Look at the residual plots. Is there evidence of heteroscedasticity (funnel shape)? What does this mean?

Project 4: Hyperparameter Tuning

Key Concepts

Term	Definition
Overfitting	Model performs well on training data but poorly on new data
Underfitting	Model is too simple to capture patterns
Cross-Validation	Split data into k folds, train on k-1, validate on 1
Grid Search	Try all combinations of hyperparameters
Learning Curve	Plot of performance vs training set size

Questions

1. Why do we need validation data separate from test data?

Answer:

2. The learning curve shows training and validation scores converging. What does this tell you about overfitting?

Answer:

Project 5: Classification Demo

Key Concepts

Term	Definition
K-NN	K-Nearest Neighbours — classify by majority vote of k neighbours
Decision Tree	Tree of if-then rules for classification
Accuracy	$(TP + TN) / (\text{Total predictions})$
Precision	$TP / (TP + FP)$ — of predicted positives, how many are correct
Recall	$TP / (TP + FN)$ — of actual positives, how many are found

Questions

1. When $k=1$ in K-NN, what happens? Why is accuracy lower than $k=7$?

Answer:

2. In the decision tree, which feature is most important for predicting house price category?

Answer:

Project 6: Ethics & Bias

Key Concepts

Term	Definition
Algorithmic Bias	Systematic, unfair treatment of certain groups
Demographic Parity	Predictions should be similar across demographic groups
Accountability	Who is responsible for model decisions?
Carbon Footprint	CO ₂ emitted during model training

Questions

1. The model overprices Northern California and underprices Southern California. Is this bias or just accurate modeling?

Answer:

2. The data includes precise latitude/longitude. What privacy concerns does this raise?

Answer:

Project 7: Customer Churn (Extension)

Key Concepts

Term	Definition
Confusion Matrix	TP / FP / TN / FN in a table
ROC Curve	True Positive Rate vs False Positive Rate at various thresholds
AUC	Area Under the ROC Curve — measure of model quality
F1 Score	Harmonic mean of Precision and Recall

Questions

1. Look at the confusion matrices. Which model does better at finding customers who will churn?

Answer:

2. The ROC AUC for the Decision Tree is ~0.82. A perfect classifier has AUC=1.0, random has AUC=0.5. Is 0.82 good?

Answer:

Project 8: Customer Segmentation (Extension)

Key Concepts

Term	Definition
K-Means	Partitions data into K clusters based on distance
Elbow Method	Plot inertia vs K, find the "elbow"
Silhouette Score	How similar a point is to its own cluster vs others
DBSCAN	Density-based clustering — finds arbitrary shapes
Dendrogram	Tree diagram of hierarchical clustering

Questions

1. Based on the Elbow Method and Silhouette analysis, what is the optimal K? Why?

Answer:

2. What are the 5 customer segments found by K-Means? Describe each in one sentence.

Answer:

Project 9: Market Basket (Extension)

Key Concepts

Term	Definition
Support	$P(A \cap B)$ — frequency of both items appearing together
Confidence	$P(B$
Lift	$P(B$
Apriori	Algorithm that finds frequent itemsets efficiently

Questions

1. The rule "TEA TOWELS 2pk $\rightarrow$ TEA TOWELS 3pk" has Lift=2.32. Explain what this means in plain English.

Answer:

2. If Lift=0.8, what does it tell you? Would you recommend these items be placed together?

Answer:

Project 10: ANN vs CNN (Extension)

Key Concepts

Term	Definition
ANN	Artificial Neural Network — fully connected layers
CNN	Convolutional Neural Network — uses convolution filters
Convolution	Sliding window that detects patterns
Pooling	Downsampling to reduce spatial dimensions
Weight Sharing	Same filter applied to all locations

Questions

1. The ANN flattens a 28×28 image into 784 pixels. What spatial information is lost?

Answer:

2. CNN has fewer parameters (~25K) than ANN (~109K) but achieves higher accuracy. How is this possible?

Answer:



Quick Reference Card

Supervised Learning Algorithms

K-NN	Both	Intuitive, no training	Slow on large data
ANN	Both	Flexible, powerful	Needs lots of data

Model Evaluation Metrics

RMSE	$[0, \infty)$	Lower = better	Regression
Precision	$[0, 1]$	$\text{Precision} > 0.8$	Minimize false positives



Data Science Workflow

1. Define Problem
↓
2. Collect Data ← All projects use REAL data
↓
3. Clean & Preprocess ← Project 2
↓
4. Explore (EDA) ← Project 1
↓
5. Feature Engineering
↓
6. Model Selection ← Project 10
↓
7. Train & Validate ← Project 3, 4
↓
8. Evaluate ← Project 5, 7
↓
9. Deploy
↓
10. Monitor (Ethics) ← Project 6

End of Workbook · Victor 🧠